FLOODGUARD NANGKA RIVER

Complete Defense Record and Evidence Guide

Final operational-model rationale | transparent OLS MLR time-series forecasting

Item	Verified position
Final model	Candidate 4 (C4)
Equation inputs	Nangka WL(t-1), Mt. Boso-Boso Rain(t-1)
Method	OLS Multiple Linear Regression; exact-calendar lags; past inputs only
Decision status	MODEL_SELECTION_PASS

Plain-language answer: C4 uses yesterday's Nangka level and yesterday's Boso-Boso rainfall to forecast the next Nangka level. It won the frozen overall validation tournament, but it is not claimed to win every type of extreme event.

How to use this record

- Sections 1-5 explain the candidate-selection process.
- Sections 6-8 explain validation, persistence, event evidence, and runtime safety.
- Section 9 gives ready-to-say panel answers.

1. What the Model Does

Nangka is a station-specific one-day-ahead water-level forecasting component. It uses transparent OLS MLR, past-day inputs only, and a deterministic persistence fallback if rainfall telemetry is missing.

Question	Answer
Forecast horizon	One calendar day ahead, using only prior-day information
Method boundary	Transparent OLS MLR time series; no AI, black-box model, Ridge, or hidden regularization
Lag rule	Exact calendar dates, not the preceding spreadsheet row
Missing-data rule	No zero imputation, interpolation, forward-fill, or backward-fill
Safety boundary	Forecast supports monitoring. Live observations and official warning protocols remain the safety authority.

Why it is a time-series MLR model

C4 has two predictors: the prior Nangka water level and prior Boso-Boso rainfall. The water-level lag makes it time-series forecasting; K = 2 satisfies the required Multiple Linear Regression form.

2. The Frozen Selection Process

Candidates were defined before validation performance was used to select the operational model. The selection procedure therefore evaluates generalization on later chronological dates rather than choosing a model because it looked good on a known storm.

Step	Rule
1. Structure	Candidate must use past-day predictors only and meet the project MLR requirement.
2. Reliability	VIF must be <= 5.0 and required slopes must pass training-sample HC3 robust inference.
3. Validation	Fit on earlier chronology and compare frozen candidates on later unseen chronology.
4. Fair comparison	Use exact common validation dates for pairwise comparisons.
5. Deployment	Consider sensor/input availability, persistence benchmark, and safe fallback separately.

Panel line: “We selected under a pre-defined overall validation process. We did not reselect the model by looking only at a few named typhoons after the fact.”

3. Why Candidate 4 (C4) Was Chosen

C4 passed HC3/VIF requirements and won the frozen 15-candidate chronological validation tournament under the primary pairwise-MAE rule: rank 1 with 14 out of 14 pairwise wins on exact common validation dates. It was chosen for best overall generalization under that pre-set rule, not because it won every extreme subset.

Selection dimension	Evidence
Model form	K = 2 OLS MLR time series: Nangka WL(t-1) plus Boso-Boso rain(t-1).
HC3 / VIF	Max VIF = 1.5957; upstream-rain HC3 p = 1.93e-6 in the final refit evidence.
Validation basis	Rank 1 of 15; 14/14 pairwise wins under the frozen exact-common-date validation MAE rule.
Operational basis	Rain missing triggers true persistence; prior Nangka WL missing returns unavailable.

What this decision does not claim

• C4 did not have the lowest RMSE across all 15 candidates.

• C4 did not win every extreme-event subset against C1 and C6.

• C4 does not guarantee flood-peak prediction or replace official warnings.

4. HC3 and VIF in Simple Terms

OLS creates the equation. HC3 is a cautious uncertainty check: it asks whether a predictor still appears dependable when ordinary days and unusual large-error days do not behave equally. VIF checks whether two inputs are so similar that the separate coefficients become unstable.

Check	Question it answers	Effect
HC3	“Can we still trust this predictor after a cautious error calculation?”	Does not change the fitted OLS prediction equation; it changes the uncertainty and eligibility evidence.
VIF	“Are two predictors too redundant?”	Prevents unstable coefficients caused by overlapping inputs.

HC3 is a quality gate, not the objective function. It does not make a forecast weaker or stronger; the equation is still ordinary least squares.

5. Chronological Validation and Persistence

C4 was selected by the frozen tournament rule, which prioritizes validation MAE and exact-common-date pairwise performance after screening. RMSE and persistence results are shown separately so no trade-off is hidden.

Model / benchmark	N	MAE (m)	RMSE (m)	Meaning
C4 selected	318	0.143768	0.384644	Tournament rank 1; 14/14 pairwise wins
True persistence	318	0.126774	0.440142	Lower MAE; higher RMSE
C5 (context)	candidate-specific	not selection basis	0.380929	Lower RMSE than C4; does not overturn frozen MAE rule

Why persistence is shown separately

True persistence is a deterministic benchmark: prediction(t) = observed water level(t-1). It is not the same as a fitted AR1 regression, which has an estimated intercept and slope.

> **On C4's identical validation dates, persistence had lower MAE (0.126774 m versus 0.143768 m), while C4 had lower RMSE (0.384644 m versus 0.440142 m). C4 was selected by the pre-defined overall tournament rule, not by hiding persistence.**

6. Extreme-Day and Event Evidence

The package event comparisons cover C4, C6, C1, and persistence. They are subset-specific safety evidence, not an all-15-candidate extreme-event selection tournament.

Subset	Best among C4/C6/C1/persistence	C4 position
Typhoon Carina	C4	Best
Typhoon Enteng	C1	C4 was worse than C1 and persistence
High-water days	C6	C4 not best
Rapid-change days	C1	C4 not best
Rapid-rise days	C4	Best
Operational-critical union	C6	C4 close but not best

Correct interpretation

C4 has meaningful strengths on Carina and rapid-rise days, but the evidence is mixed across subsets. The honest conclusion is that C4 is the overall tournament winner; it is not the universal extreme-event winner.

Do not say the selected model is proven best for every typhoon. The candidate is the most defensible overall operational choice under the frozen methodology; event evidence is reported as a separate safety check.

7. Deployment and Fallback

Runtime condition	Output
Nangka(t-1) and Boso-Boso rain(t-1) valid	Calculate Candidate 4 OLS MLR forecast.
Nangka(t-1) valid; Boso-Boso rain missing	Use true persistence: prediction = Nangka(t-1).
Nangka(t-1) missing	Return forecast unavailable; no imputation.

Why this is safe

• The model never invents a missing prior water-level value.

• Persistence is an explicit deterministic fallback, not a hidden substitute model.

• The system must show the forecast source/fallback state so users know what was used.

• A forecast supplements, never replaces, live monitoring and official warnings.

8. What is Recorded as Proof

Evidence item	What it proves
Frozen candidate definitions and checksum	Candidates were specified before selection.
Model-ready data and exact-calendar-lag audit	No row-shift lag leakage; missing values retained.
OLS, HC3, and VIF tables	Equation, uncertainty, and collinearity evidence.
Validation predictions and pairwise comparisons	Later-date performance and fair common-date comparison.
Persistence and event-subset files	Benchmark trade-offs and stated extreme-day evidence.
Formula-backed Excel workbook	Visible native calculation trail with formula-error scan.
Runtime engines, parity tests, manifest, checksum	Deployment equation matches the audited model and is reproducible.

The package is the source of truth. This defense record summarizes that evidence and never replaces the CSV, workbook, or reproducibility trail.

9. Panel Q&A;: Say It Simply

Likely question	Answer
Why this model?	C4 passed HC3/VIF and won the frozen 15-candidate chronological MAE tournament on exact common dates.
Did HC3 pick it by itself?	No. HC3 and VIF were gates. The final choice then followed frozen chronological validation and operational rules.
Is it best in every typhoon?	No. We do not make that claim. Extreme-day results are separate, subset-specific evidence.
Why not select using only typhoon days?	That would be hindsight selection on a small, known set of events and could overfit those events.
What happens when an input is missing?	If rain is missing, use true persistence. If prior Nangka level is missing, return unavailable - never impute.
What protects the public?	Live measurements, official warning protocols, and human/agency decision-making remain the safety authority.

Closing defense position: “Our model is not presented as perfect for every storm. It is the most defensible station-specific OLS MLR operational model under a frozen, transparent, reproducible selection process.”